ASSIGNMENT

LAB 3:

TASK 1

PROGRAM:

```
#include <iostream>
using namespace std;
int main ()
{
    int age;
    double Pay;
    char section;
    cout<<"Enter age, Pay and section: ";
    cin>>age >> Pay >>section;
    cout << "Age: " <<age <<endl;
    cout << "Pay: " <<Pay <<endl;
    cout <<section: " << section << endl;
    return 0;
}
```

OUTPUT

```
Age : 20
Pay : 15000.5
Section : A
```

Coding C++

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RUN MENU

```cpp
#include <iostream>
using namespace std;
int main()
{
    int age;
    double pay;
    char section;
    cout << "Enter age, pay and section: ";
    cin >> age >> pay >> section;
    cout << "Age: " << age << endl;
    cout << "Pay: " << pay << endl;
    cout << "Section: " << section << endl;
    return 0;
}
```

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Compile Result

```
Enter age, pay and section: 20  15000.50 A
Age: 20
Pay: 15000.5
Section: A

[Process completed - press Enter]
```

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TASK 2:

28 / 4 - 2 = 5

6 + 12 * 2 - 8 = 22

4 + 8 * 2 = 20

6 + 17 % 3 - 2 = 6

2 + 22 * (9 - 7) = 46

(8+7) * 2 = 30

(16 + 7) % 2 - 1 = 0

12 / (10 - 6) = 3 → (19-3) * (2+2) / 4 = 16

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code

```
28 / 4 - 2 = 5
6 + 12 * 2 - 8 = 22
4 + 8 * 2 = 20
6 + 17 % 3 - 2 = 6
2 + 22 * (9 - 7) = 46
(8 + 7) * 2 = 30
(16 + 7) % 2 - 1 = 0
12 / (10 - 6) = 3
(19 - 3) * (2 + 2) / 4 = 16
```

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TASK 3:

Correct answer is B

REASON: In C++, when two different data types are used in an arithmelic operation, the smaller data type is automatically promoted to the larger data type. This is called type promotion.

FOR THE EXPRESSION MASS_UNITS

1. units is an int and mass is a float
2. int is smaller than float, so units will be promoted to a float.
3. mass remains a float.
4. Therefore, the result of mass*units will be a float.

Then for the full statement weight = mass*units

5. The float result is then promoted to a double because weight is a double variable.

So: Option B is 100% correct for the mass_units part.

Option C describes the final value after it is assigned to weight.

9:10 • 71%
Meta AI

Here is the reason for **Task 3** in English:

Correct Answer is B

Reason:

In C++, when two different data types are used in an arithmetic operation, the smaller data type is automatically promoted to the larger data type. This is called **Type Promotion**.

For the expression `mass _ units`:

1. `units` is an `int` and `mass` is a `float`.
2. `int` is smaller than `float`, so `units` will be **promoted to a float**.
3. `mass` remains a `float`.
4. Therefore, the result of `mass _ units` will be a **float**.

Then for the full statement `weight = mass _ units`:

5. The `float` result is then promoted to a

Message

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Then for the full statement `weight = mass_units`:

5. The `float` result is then promoted to a `double` because `weight` is a `double` variable.

So:

Option B is 100% correct for the `mass_units` part.

Option C describes the final value after it is assigned to `weight`.

9:09 pm

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TASK 4:

View Code

```
int a, b = 2;
float c = 4.2;
a = b * c;
```

CORRECT ANSWER IS B. 8

Reason:

1. First the expression b - c is evaluated 2 - 4.2 = 8.4
2. The result 8.4 is of type float, because one operand (c) is a float.
3. a is defined as int. When a float value is assigned to an int variable, implicit type conversion (demotion) occurs.
4. The fractional part is truncated (remo not rounded.
5. Therefore, 8.4 becomes 8 and is store in a.

Question:

Cpp code
View code

```cpp
int a, b = 2;
float c = 4.2;
a = b * c;
```

Correct Answer: B. 8

Reason in English:

1. First, the expression `b * c` is evaluated: `2 * 4.2 = 8.4`
2. The result `8.4` is of type `float`, because one operand (`c`) is a float.
3. `a` is defined as an `int`. When a float value is assigned to an int variable, implicit type conversion (demotion) occurs.
4. The fractional part is truncated (removed), not rounded.
5. Therefore, `8.4` becomes `8` and is stored in `a`.

9:14 pm

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TASK 5:

Code

UnitsEach = static_cast<float>(qty)

REASON:

qty ko float / division, b double mein cast karne se division float division ban jaye ga aur decimal result ayega, jaise 7/2 = 3.5 ki jagah 3 nhi hoga.

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code

View code

```
unitsEach = static_cast<float>(qty)
```

Reason: qty ko float/double mein cast karne se division float division ban jayega aur decimal result aayega, jaise 7 / 2 = 3.5 ki jagah 3 nahi aayega.

9:17 pm

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TASK 6:

```
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;
int main () {
    srand (time (0));
int num1 = rand ( ) % 900 + 100;
int num2= rand ( ) % 900 + 100;
int sum = num1 + num2;
cout <<"     " <<num1 << endl;
cout <<"+    " <<num2 << endl;
cout <<"---------" <<endl;
cout <<"\n Press Enter to see the answer....";
cout <<"\n   " <<num1 <<endl;
cout <<"+    " << num2 <<endl;
cout <<"--------" <<endl;
cout <<"    " << sum << endl;
    return 0;
}
```

OUTPUT

```
 119
+144
-------
 263
```

Coding C++

Nothing changed

RUN MENU

```cpp
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;

int main()
{
    srand(time(0));
    int num1 = rand() % 900 + 100;
    int num2 = rand() % 900 + 100;
    int sum = num1 + num2;

    cout << "  " << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "--------" << endl;

    cout << "\nPress Enter to see the answer...";
    cin.get();

    cout << "\n  " << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "--------" << endl;
    cout << "  " << sum << endl;

    return 0;
}
```

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Compile Result

```
  119
+ 144
--------

Press Enter to see the answer...

  119
+ 144
--------
  263

[Process completed - press Enter]
```

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HOME TASK 1:

PROGRAM 1

```
#include <iostream>
using namespace std;
int main ()
{
   double number1, number2, sum;
   cout << "Enter a number: ";
   cin >> number1;
  cout << "Enter another number: ";
   cin >> number2;
   sum = number1 + number2;
   cout << "The sum of the two numbers
  is " << sum;
   return 0;
}
```

OUTPUT

```
25
30
Sum = 55
```

PROGRAM 2

```
#include <iostream>
using namespace std;
int main ()
{
   int number1, number2;
  float quotient;
  cout << "Enter two numbers and I will divide
  \n";
  cout << "the first by the second for you
  \n";
  cin >> number1 >> number2;
  quotient = static_cast<float>(number1) / number2;
   cout << quotient;
     return 0;
}
```

OUTPUT

```
7 / 2 = 3.5
```

Coding C++

Auto saved at 21:34:18

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    double number1, number2, sum;
    cout << "Enter a number: ";
    cin >> number1;
    cout << "Enter another number: ";
    cin >> number2;
    sum = number1 + number2;
    cout << "The sum of the two numbers is " << sum;
    return 0;
}
```

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Compile Result

```
Enter a number: 25
Enter another number: 30
The sum of the two numbers is 55
[Process completed - press Enter]
```

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Coding C++

Auto saved at 21:36:09

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int number1, number2;
    float quotient;
    cout << "Enter two numbers and I will divide\n";
    cout << "the first by the second for you.\n";
    cin >> number1 >> number2;
    quotient = static_cast<float>(number1) / number2;
    cout << quotient;
    return 0;
}
```

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Compile Result

```
Enter two numbers and I will divide
the first by the second for you.
7 2
3.5
[Process completed - press Enter]
```

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TASK 2:

```
#include <iostream>
using namespace std;
int main ()
{
    int val1 = 28;
    int val2 = 32;
    int val3 = 37;
    int val4 = 24;
    int val5 = 33;
int sum = val1 + val2 + val3 + val4 + val5;
double average = sum / 5.0;
cout << " The average is : " << average << endl;
    return 0;
}
```

OUTPUT

Average 30.8

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Coding C++

Auto saved at 21:40:41

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int val1 = 28;
    int val2 = 32;
    int val3 = 37;
    int val4 = 24;
    int val5 = 33;

    int sum = val1 + val2 + val3 + val4 + val5;

    double average = sum / 5.0;

    cout << "The average is: " << average << endl;

    return 0;
}
```

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Compile Result

```
The average is: 30.8

[Process completed - press Enter]
```

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